Shaping the Future of Power Electronics

By delivering the Most Efficient Easy-to-use transistor



CGD Founders and senior team: excellence from Cambridge University



Dr. Giorgia Longobardi

Founder and CEO

- 11 years experience in Gallium Nitride Power devices.
- Product-oriented research project manager with top semiconductor companies: NXP (2010-2013), Infineon (2014-2017), Vishay (2014-2017).
- Leader of the GaN Power devices team at the Engineering Department of University of Cambridge.
- Inventor of 12 patent applications.



Prof. Florin Udrea

Founder and CTO

- Professor of Power semiconductor devices at University of Cambridge (Si, SiC, GaN, Diamond).
- Inventor of more than 150 patent applications.
- Consultant for: Toyota, Denso, NXP, Infineon, Vishay, On Semi, ABB.
- Founder of CamSemi (Si power ICs) sold to Power Integration.
- Founder of Cambridge CCS (Sensors) sold to AMS.
- Founder of Camutronix (Si power devices).



Zahid Ansari

VP of Operations

- VP of Operations, Director of Product Engineering and Acting VP of Business Development at CamSemi.
- Responsible for enabling CamSemi's supply chain to deliver over 500,000 IC's per day and over 1 billion IC's in total.
- 14 years of experience in technical and operations roles in Power Electronics.



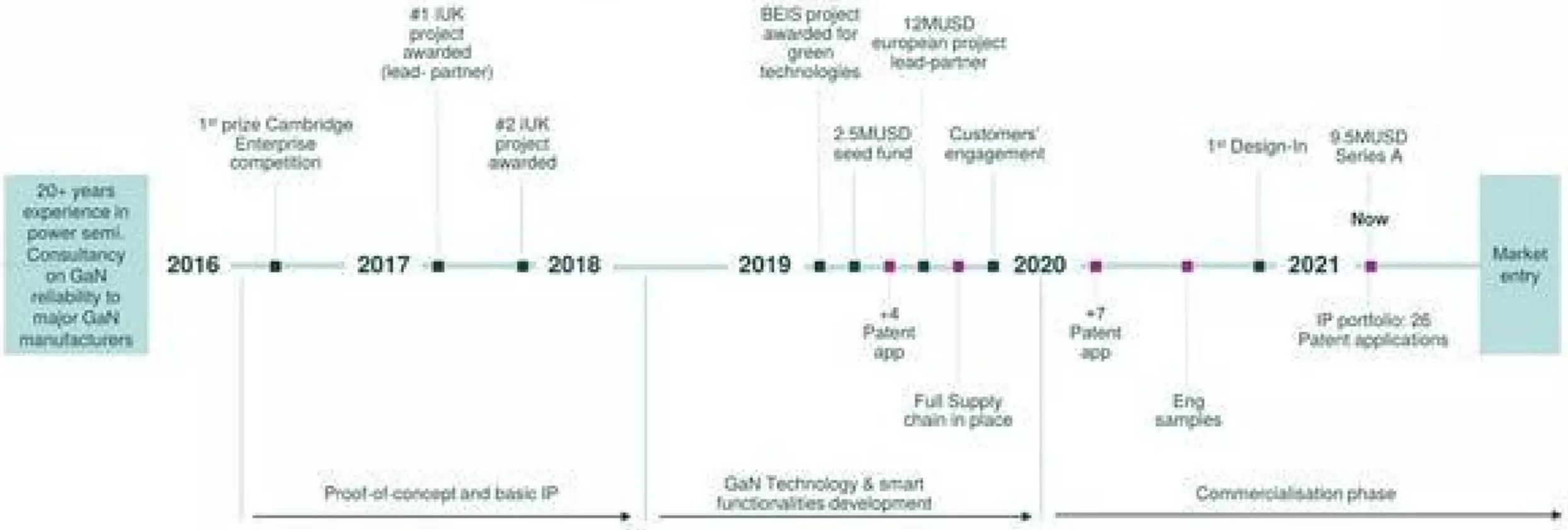
Andrea Bricconi

VP of Business development

- 20+ years experience in power semiconductors industry.
- Led product engineering, product development and marketing groups in International Rectifier and Vishay.
- Co-started and managed the GaN program at Infineon with focus on business development for consumer, industrial and automotive markets.
- Focus on customers and on bringing new technologies to the market.

CGD timeline and milestones

Extremely successful in securing public funds and committed to overdeliver on our milestones



Power devices at the heart

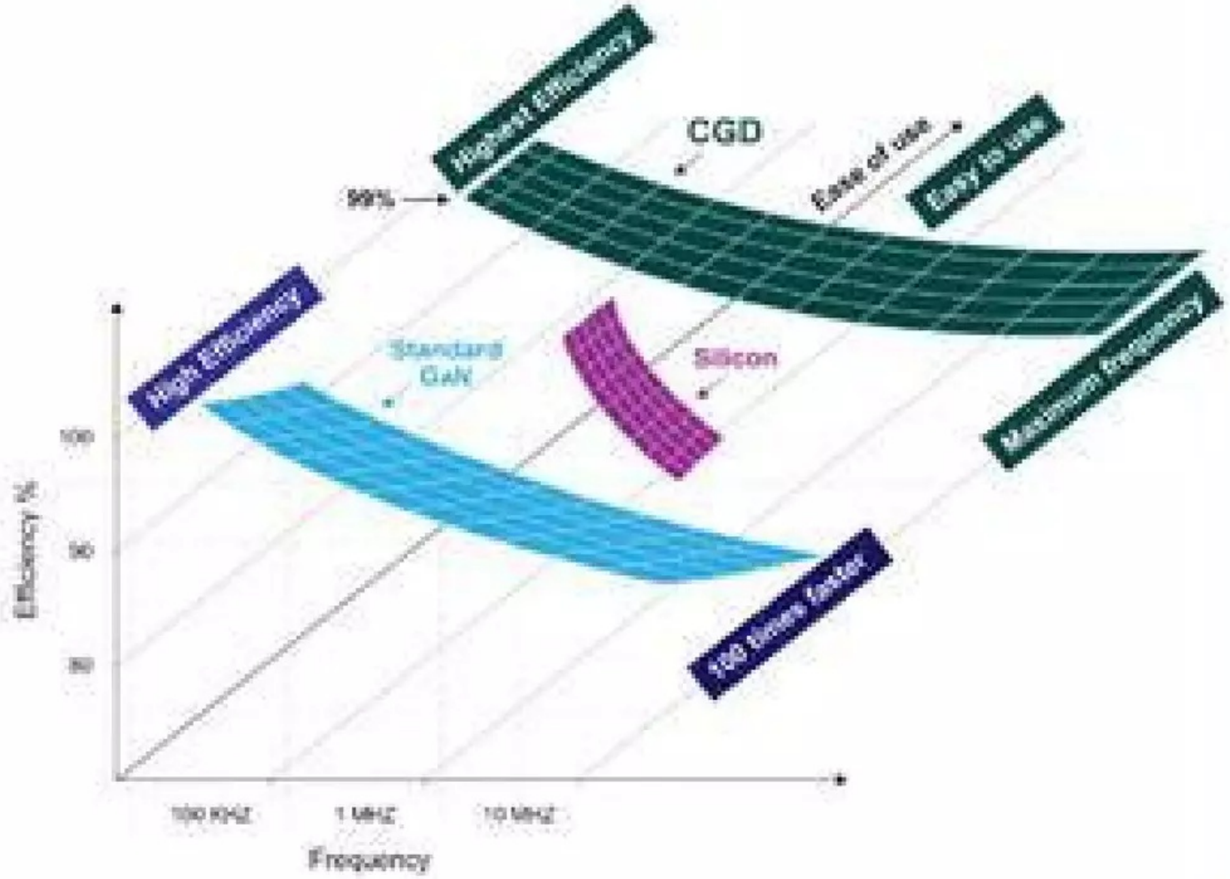
>80% of the world's electricity passes through a Power Electronics circuit…



…and is controlled by Power Devices



What if…CGD?



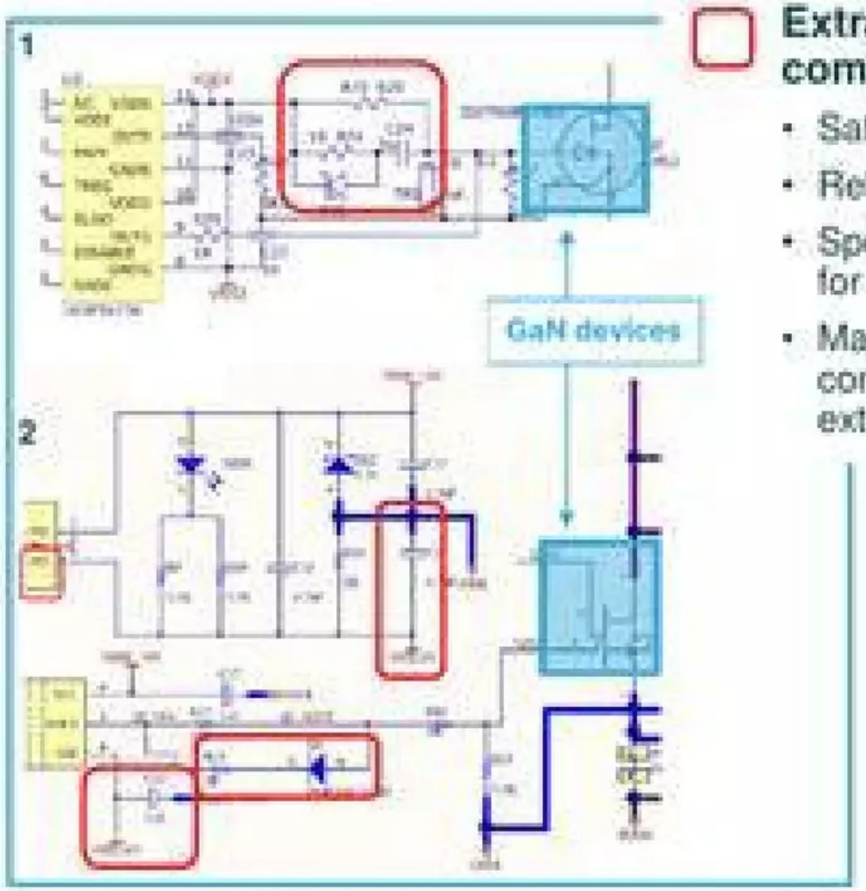
CGD technology:
The best of silicon with the performance of GaN

- ✔ Reliable
- ✔ Safe
- ✔ Freedom of design
- ✔ Not as hot, more reliable, no cooling components
- ✔ Hundreds of TWh per year saved
- ✔ Billions of Energy bills saved
- ✔ Millions of tons of CO_2 saved
- ✔ Smaller, more compact circuit
- ✔ More power for the same volume
- ✔ Maximum Efficiency

CGD device: a self-protected chip, easy to be interfaced

Examples from competition

Additional devices needed to make GaN operating

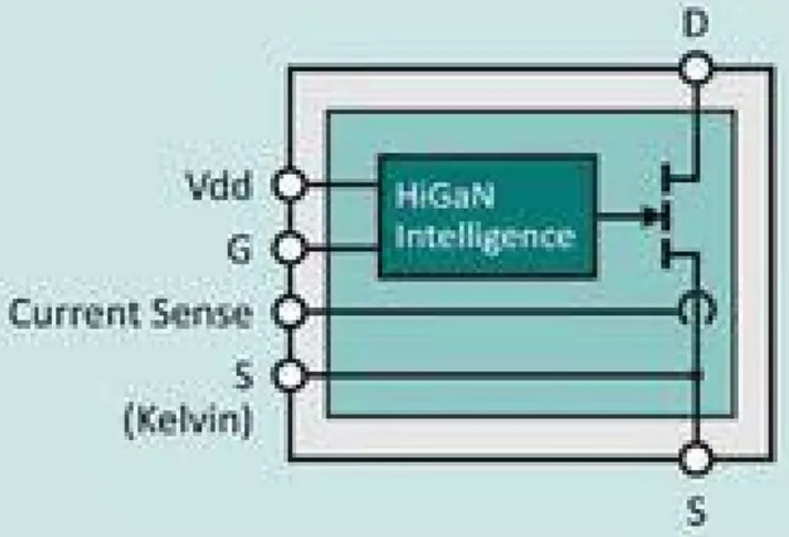


Extra components for:

- Safety
- Reliability
- Specific gate driver for correct functioning
- Maximum efficiency compromised by extra components

Cambridge GaN Devices

No extra components are needed



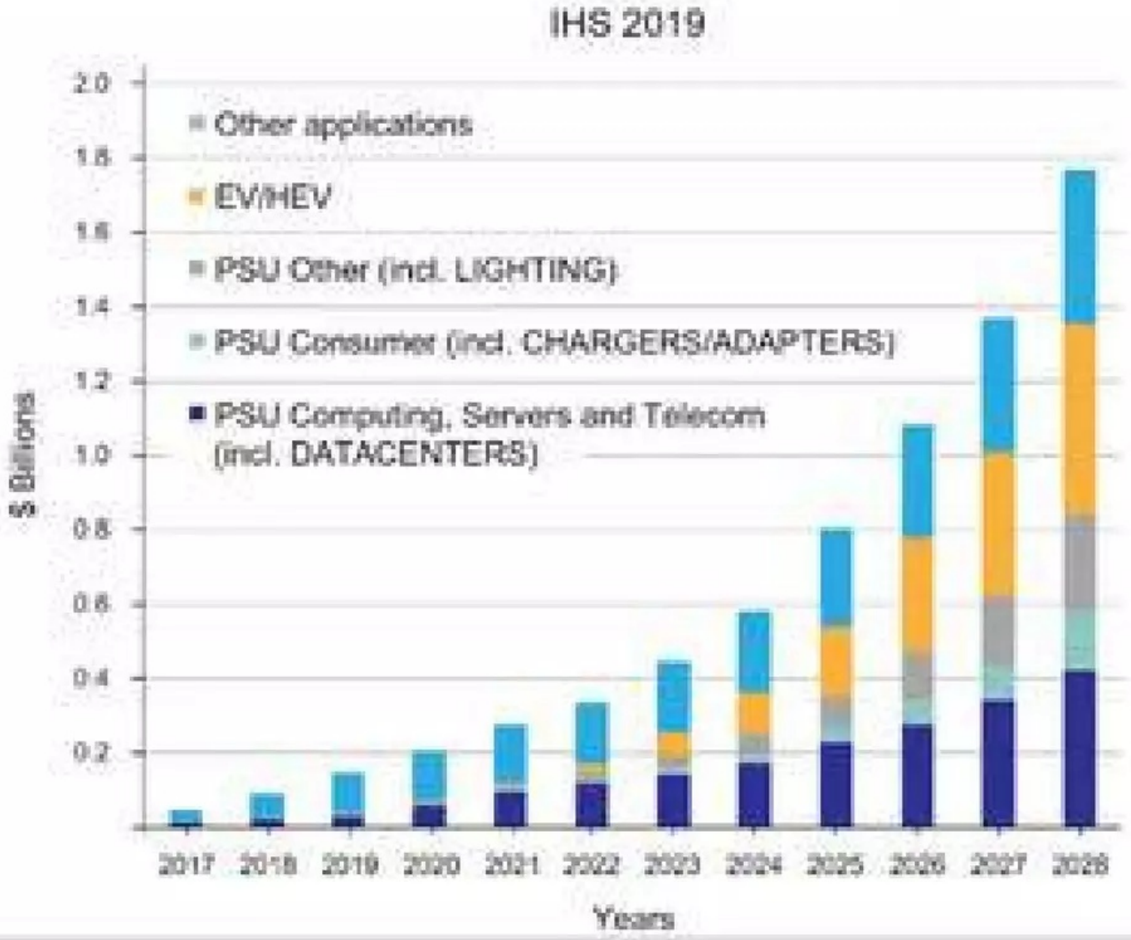
HiGaN intelligence = **CGD on-chip integrated IC solution remove the needs for extra components**

Market opportunity

CGD addresses the fast growing GaN power devices market predicted to reach over $1bn in 2026

Power semiconductor market

TAM
$45 Billions



Power supplies and automotive are the largest markets

The markets we operate in

Consumer electronics: Power supplies



Reduced weight

More efficient

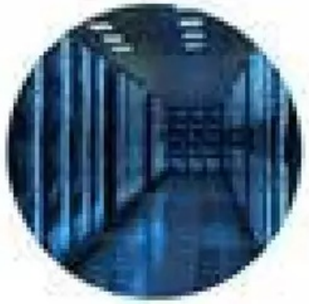
Industrial: LED drivers and power supplies



Volume reduction

More efficient

Data centres: Power supplies



OPEX reduction by 10% or more

CAPEX reduction up to 10%

Automotive, electric vehicles: On-Board-Chargers (OBC)



Faster charging

50% smaller OBCs and DCDC converters

WHY NOW?

A compact and green solution

Hi-tech horticulture for national supply chain.

More and more connected

Huge demand for audio/video streaming and online services

Towards a green world

Electrifications of cars is happening

GaNext: an example of international cooperation

CGD is leading the European funded project aimed at developing GaN-based modules for low and high power applications

GaNext

13 partners

3 countries

12.5 million $

Cambridge GaN Devices

Penta EUREKA innovation across borders

advICo SiC	Besi (NL) Packaging Technology	CSA Catapult (UK) Application testing	TU Eindhoven (NL) Power Electronics	Fraunhofer IMS (DE) SiC IC	Maccon (DE) Motor Drive
Infineon Technologies (DE) Gate Driver	Lyra Electronics (UK) On Board Charger	Neways Technologies (NL) PV Inverter	Signify (NL) LED driver	Sumida (DE) Magnetic components	TU Dortmund (DE) Power Electronics

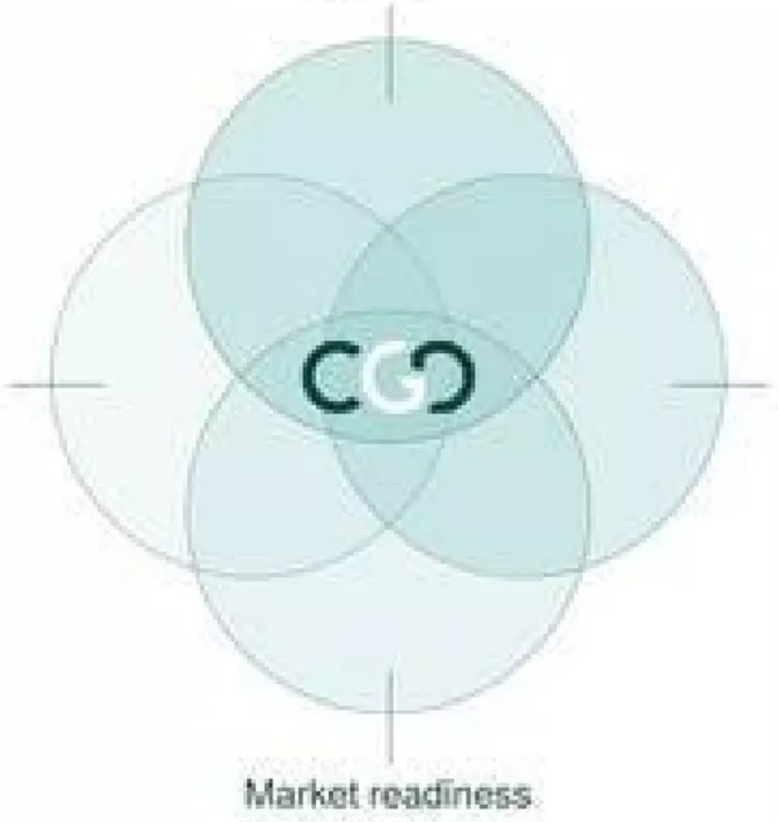
CGD gathered professionals from the industry

Where innovation, unique proposition, market readiness and experience meet

Experienced team with track record in power and fabless semiconductor business

Unique proposition

University of Cambridge
Brand Cutting-edge research
Pool of talents



Market readiness

Cambridge GaN Devices

First Floor, Deanland House, 160 Cowley Road, Cambridge CB4 0DL, UK

Giorgia Longobardi - CEO
e-mail: giorgia.longobardi@camgandevices.com
mobile: +44 7446 332277

Andrea Bricconi - VP Business development
e-mail: andrea.bricconi@camgandevices.com
mobile: +49 173 2410796

